GENERAL ANESTHETICS-I

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DIKIOHS

FOR PATIENTS UNDERGOING SURGICAL OR MEDICAL PROCEDURES DIFFERENT LEVEL OF SEDATION CAN PROVIDE IMPORTANT BENEFITS TO FACILITATE PROCEDURAL INTERVENTIONS

- SEDATION
- ANXIOLYSIS
- ANALGESIA
- AMNESIA (LACK OF AWEARNESS)
- SKELETAL MUSCLE RELAXATION
- SUPRESS UNDESIRABLE REFLEXES

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PREANESTHTIC MEDICATION

PROVIDE ANXIOLYSIS AND ANALGESIA FACILITATE SMOOTH INDUCTION OF ANESTHESIA, LOWER REQUIRED ANESTHETIC DOSES

- H_2 BLOCKERS (FAMOTIDINE, RANITIDINE)
- BENZODIAZIPINES (MIDAZOLAM, DIAZEPAM)
- NONOPIOIDS: ACETAMINOPHEN, CELECOXIB)
- OIPIODS (FETANYL)
- ANTIHISTAMINE H_1 BLOCKERS(DIPHENHYDRAMINE)
- ANTIEMETICS (ONDANSETRON)
- ANTIMUSCARINIC(GLYCOPYRROLATE)

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GENERAL ANESTHESIA

REVERSIBLE STATE OF CNS DEPRESSION CAUSING LOSS OF RESPONSE TO AND PERCEPTION OF STIMULI

PHYSIOLOGIC STATE INDUCED BY GENERAL ANESTHETICS INCLUDES

- ANALGESIA
- AMNESIA
- LOSS OF CONSCIOUSNESS
- INHIBITION OF SENSORY & AUTONOMIC REFLEXES
- SKELETAL MUSCLE RELAXATION

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STAGES OF ANESTHESIA

THREE STAGES

1. INDUCTION

- **TIME FROM ADMINISTRATION OF A POTENT ANESTHETIC TO DEVELOPMENT OF EFFECTIVE ANESTHESIA**
- **DEPENDS ON HOW FAST EFFECTIVE CONCENTRATIONS OF ANESTHETICS REACH THE BRAIN**
- **NORMALLY (ADULTS)INDUCED WITH IV AGENT E.G. PROPOFOL**
- **PATIENT BECOME UNCONCIOUS IN 30 TO 40 SECONDS**
- **FOR CHILDREN NONPUNGENT SEVOFLURANE IS INHALED**
- **OFTEN, I/V ROCURONIUM, VACURONIUM OR SUCCINYLCHOLINE (NM BLOCKRS) IS ADMINISTERED TO FACILITATE ENDOTRACHEAL INTUBATION**

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QUESTION

A 25-year-old man is brought to the operating room. The anesthetist starts administering an inhalational anesthetic and the patient begins to lose consciousness.

A. Surgical anesthesia
B. Induction
C. Recovery
D. Medullary paralysis

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2. MAINTENANCE

- PROVIDE SUSTAIN PERIOD OF GENERAL ANESTHESIA
- VITAL SIGNS AND RESPONSE TO STIMULI ARE OBSERVED
- PROVIDED WITH VOLATILE ANESTHETICS
- FENTANYL IS ALSO ADMINISTERED ALONG WITH INHALATION ANESTHTICS

3. RECOVERY

- TIME FROM DISCONTINUATION OF ANESTHETICS UNTIL CONCIOUSNESS AND PROTECTIVE REFLEXES RETURN
- DEPENDS ON HOW FAST THE ANESTHETIC DIFFUSES FROM THE BRAIN (REDISTRIBUTIION)
- PATIENT MONITORED FOR BP, HR, RR AND PROTECTIVE REFLEXES

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DEPTH OF ANESTHESIA

DEGREE TO WHICH CNS IS DEPRESSED

I. STAGE OF ANALGESIA (LOSS OF PAIN)

II. STAGE OF EXCITEMENT (DISPLAY DILIRIUM)

III. STAGE OF SURGICAL ANESTHESIA (LOSS OF MUSCLE TONE AND RELEXES)

IV. STAGE OF MEDULLARY DEPRESSION (SEVERE DEPRESSION OF REPIRATORY AND VASOMOTOR CENTER)

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TYPES OF GENERAL ANESTHETICS

I/V ANESTHETICS

1) BARBITURATES

E.G. THIOPENTAL, METHOHEXITAL

2) BENZODIAZEPINES

E.G. MIDAZOLAM, DIAZEPAM

3) PROPOFOL

4) KETAMNE

5) OPIOD ANALGESIC

E.G. MORPHINE, FENTANYL, SUFENTANIL, ALFENTANIL, REMIFENTANIL

6) MISCELLANEOUS SEDATIVE-HYPNOTICS

E.G. ETOMIDATE, DEXMEDETOMIDINE

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INHALED ANETHETICS

USED PRIMARILY FOR MAINTENANCE OF ANESTHESIA AFTER ADMINISTRATION OF AN IV DRUG.

VOLATILE INHALED ANESTHETICS

E.G - HALOTHAN

- ISOFLURANE
- SEVOFLURANE
- DESFLURANE
- ISOFLURANE

GAS

E.G. NITROUS OXIDE

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BALANCED ANESTHESIA

QUESTION

During a lecture, the professor asks students to identify the inhalational anesthetic that is not a halogenated hydrocarbon.

A. Halothane

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B. Isoflurane

C. Sevoflurane

D. Nitrous oxide

INHALATION ANESTHETICS

- HALOGINATED HYDROCARBON AND NITROUS OXIDE
- MAINTENANCE OF ANESTHESIA
- STEEP DRC
- NARROW TI
- NO ANTAGONIST AVAILABLE
- NARROW GAP BETWEEN ELICITING GA TO CP COLLAPSE
- NONFLAMMABLE
- NONEXPLOSIVE
- DECREASES CEREBRAL VASCULAR RESISTANCE
- INCREASES BLOOD FLOW TO BRAIN
- INCREASES BRAIN PERFUSION
- CAUSE BROCHODILATION
- DECREASE BOTH REPIRATORY DRIVE

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POTENCY OF INHALATION ANESTHETICS

- MINIMUM ALVEOLAR CONCENTRATION: END-TIDAL CONC. OF INHALED ANESTHETICS NEEDED TO ELIMINATE MOVEMENT IN 50% OF PATIENTS STIMULATED BY INCISION
- MAC IS SMALL FOR POTENT ANESTHETICS E.G. ISOFLURANE
- MAC IS LARGE FOR LESS POTENT ANESTHETICS E.G. NITROUS OXIDE
- MORE LIPOPHILIC AN ANESTHETIC THE LOWER THE CONCENTRATION NEEDED TO PRODUCE ANESTHESIA THEREFORE HIGHER THE POTENCY

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FACTORS INCREASES THE MAC VALUE

- HYPERTHERMIA
- CHRONIC ETHANOL ABUSE
- DRUGS INCREASES CNS CATECHOLAMINES

FACTORS DECRESASES THE MAC VALUE

- INCREASED AGE
- HYPOTHERMIA
- PREGNANCY
- SEPSIS
- CONCURRENT I/V ANESTGETICS
- ALFA -2 ADRENERGIC AGONIST (CLONIDINE, DEXMEDETOMIDINE)

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QUESTION

A resident compares two inhalational anesthetics and asks which one has the lowest MAC, indicating the highest potency.

A. Nitrous oxide
B. Desflurane
D. Sevoflurane

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PARTIAL PRESSURE OF ANESTHETIC GAS ORIGINATES IN LUNGS IS THE DRIVING FORCE MOVING ANESTHETIC GAS FROM ALVEOLAR SPACE IN TO THE BLOOD STREAM WHICH TRANSPORTS THE DRUG TO THE BRAIN

ALVEOLAR WASH IN: NORMAL LUNG GAS IS REPLACED WITH INSPIRED ANESTHETIC MIXTURE

ANESTHETIC UPTAKE: UPTAKE OF GAS TO PERIPHRAL TISSUES (EXCLUDING BRAIN)

- **SOLUBILITY OF GAS IN BLOOD**
- **CARDIAC OUT PUT**
- **GRADIENT BETWEEN ALVEOLAR AND BLOOD ANESTHETIC PARTIAL PRESSURE**

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SOLUBILITY IN BLOOD

Blood Gas PC: The Ratio Of The Concentration Of Anesthetic In The Liquid (Blood) Phase To The Concentration Of The Anesthetic In The Gas Phase When The Anesthetic Is In Equilibrium Between The Two Phases

Anesthetic With Low Blood Solubility-----the Brain Reaches The Anesthetic Partial Pressure Quickly ----Faster Induction----example Is Nitrous Oxide And Desflurane

Anesthetic With High Blood Solubility---- Blood Absorbs A Large Amount Of Anesthetic-----brain Partial Pressure Rises Slowly------slower Induction ----Greater Concentration And Longer Time Required--- ---Example Is Isoflurane

Solubility In Blood: Isoflurane Greater Than Sevoflurane Greater Than Nitrous Oxide Greater Than Desflurane

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QUESTION

A patient receives INHALATION anesthetics for maintenance of anesthesia. The anesthetist explains that its blood solubility is lower compared with other volatile agents.

A. Isoflurane
B. Sevoflurane
C. Desflurane

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CARDIAC OUTPUT:

CO Inversely Correlated With Induction Time For Inhaled Anesthetics (How Much Blood Flows Thru The Lungd Each Minute)

Low CO:

(Less Blood Passes Thru The Lungs)

LONGER PERIOD OF TIME PERMITS A LARGER CONCENTRATION OF GAS TO DISSOLVE IN SLOWLY MOVING BLOODSTREAM (Less Anesthetic Is Removed From The Alveoli---alveolar Partial Pressure Rapily Ises---brain Partial Pressure Rises Quickly---faster Induction)

High CO:

(More Blood Passes Thru The Lungs)

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ALTHOUGH THE HIGH CO QUICKLY TRANSPORT THE DRUG TO THE BRAIN , A LOWER CONCENTRATION OF ANESTHETIC WITH SHORTER EXPOSURE TIME SLOWS DOWN THE RATE OF INDUCTION (More Anesthetic Is Taken Up From The Alveoli In To The Blood----alveolar Partial Pressure Rises More Slowly----less Anesthetic Reaches The Brain Initially----slower Induction)

MECHANISM OF ACTION OF GENERAL ANESTHETICS

1. Increase **Sensitivity Of Gaba –A Recceptors To Gaba (Inhibitory Nt**) That Results In Chloride Influx And Hyperpolarization Of Neurons That

Decreases Neuronal Excitability &

Diminished CNS Activity

2. **Inhibition Of NMDA Receptors** (Excitatory Glutamate NT) By Nitrous Oxide And Ketamine

3. **Increase Activity Of Glycine** (Inhibitory NT)

4. **Block Excitatory Postsynaptic Currents Of Nicotinic Receptors (Inhalation Anesthetics)**

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HALOTHANE

- PROTOTYPE
- RAPID INDUCTION
- QUICK RECOVERY
- UNSAFE ADVERSE EFFECTS PROFILE
- POTENNT ANESTHETIC
- RELATIVELY WEAK ANALGESIC
- POTENT BRONCHODILATOR
- RELAXIS SKELETAL AND UTERINE SMOOTH MUSCLE
- USED IN OBSTETRICS
- NOT HEPATOTOXIC IN CHILDREN
- SUITABLE IN PEDIATRIC PATIENTS
- PLEASANT ODOR

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- METABOLIZED TO TISSUE TOXIC HYDROCARBONS (TRIFLOUROETHANOL AND BROMIDE)
- TOXIC REACTIONS: FEVER, ANOREXIA, NAUSEA, VOMITING, HEPATITIS (LOW INCIDENCE 1: 10,000)
- AVOID ADMINISTRATION OF HALTHONE AT INTERVALS OF LESS THAN 2-3 WEEKS
- VAGOMIMETIC CAUSES BRADYCARDIA
- CAUSES CARDIAC ARRYTHMIAS
- CAUSES MALIGNANT HYPERTHERMIA

(Life Threatening Condition With Hyperpyrexia, Circulatory Collapse, Dantroline Is Given)

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ISOFLURANE

- UNDERGO LITTLE METABOLISM
- LESS TOXIC TO LIVER AND KIDNEY
- PRODUCES DOSE DEPNDENT HYPOTENSION
- HYPOTENSION TREATED WITH PHENYLEPHRINE (VASOCONSTRICTOR)
- DECREASES VASCULAR RESISTANCE
- PERFUSES MAJOR TISSUES
- HAS PUNGENT ODOR
- STIMULATE RESPIRATORY REFLEXES (LARYNGIOSPASM, BREATH HOLDING, COUGHING, SALIVATION)
- NOT USED AS INDUCTION ANESTHESIA
- HIGH BLOOD SOLUBILITY
- LESS IDEAL FOR SHORT PROCEDURES
- USED WHEN COST IS A FACTOR FOR LONGER SURGERIES

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DESFLURANE

- POPULAR FOR OUT PATIENT PROCEDURES
- UNDERGO LITTLE METABOLISM
- LESS TOXIC TO LIVER AND KIDNEY
- RAPID ONSET OF ACTION
- RAPID RECOVERY
- LOW BLOOD SOLUBILITY
- POPULAR ANESTHETIC FOR SHORT PROCEDURES
- DOSE DEPENDENT DECREASE IN BP
- DECREASES VASCULAR RESISTANCE
- PERFUSES MAJOR TISSUE
- STIMULATE RESPIRATORY REFLEXES
- NOT USED FOR INHALATION INDUCTIONS
- EXPENSIVE

SEVOFLURANE

- LOW RISK OF HEPATOTOXICITY
- SOME RISK OF RENAL TOXICITY (NEPHROTOXIC)
- LESS PUNGENT
- RESPIRATORY REFLEXES INHIBITED
- RAPID ONSET OF ACTION
- SUITABLE FOR INDUCTION IN PEADS
- DOSE DEPENDENT HYPOTENSION

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QUESTION

A 5-year-old child requires mask induction before surgery. The anesthetist selects an agent because it is non-pungent and well tolerated.

A. Desflurane

B. Isoflurane

C. Sevoflurane

D. Enflurane

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QUESTION

A patient develops renal toxicity after prolonged exposure to an inhalational anesthetic.

A. Nitrous oxide

B. Sevoflurane

C. Halothane

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EFFECT	HALOTHANE	ISOFLURANE	DESFLURANE	SEVOFLURANE	NITEROUS OXIDE
MAC VALUE	0.75 %	1.4%	6%	2%	104%
POTENCY	HIGH	HIGH	LOWER THAN OTHERS	HIGH	LOW
BGPC	2.3	1.4	0.45	0.65	0.47
BLOOD SOLUBILITY	LOW	LOW	LOWER THAN OTHERS	LOW	LOW
ODOR	PLEASANT	PUNGENT	PUNGENT	PLEASANT	Pleasant
AIRWAY IRRITATION	NO	YES	YES	NO	NO
HEPATOTOX ICITY	YES	NO	NO	NO	NO

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EFFECT	HALOTHANE	ISOFLURANE	DESFLURANE	SEVOFLURANE	NITROUS OXIDE
INDUCTION	RAPID	RAPID	RAPID	RAPID	RAPID
RECOVERY	QUICK	QUICK	QUICK	QUICK	QUICK
NEPHROTOXICITY	NO	NO	NO	YES	NO
BRONCHODILATOR	YES	YES	YES	YES	NOT PRODUCES SM.MUSCLE RELAXATION
HEART RATE	BRADYCARDIA	TACHYCARDIA	TACHYCARDIA	TACHYCARDIA	
BP	DECREASE C.O DECREASE BP	DECREASE PVR DECREASE BLOOD PRESSURE	DECREASE PVR DECREASE BLOOD PRESSURE	DECREASE PVR DECREASE BLOOD PRESSURE	
ARRYTHMIAS	YES	NO	NO	NO	NO

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NITROUS OXIDE

- LAUGHING GAS
- WEAK GENERAL ANESTHETICS
- POTENT ANALGESIC
- NON-IRRITATING
- NOT ABLE TO CREATE A STATE OF GA
- FREQUENTLY USED AT CONCENTRATIONS OF 30% TO 50% IN COMBINATION WITH OXYGEN TO CREATE MODERATE SEDATION PARTCULARLY IN DENTISTRY.
- POOR BLOOD SOLUBILITY
- NOT DEPRESS RESPIRATION
- NOT PRODUCE MUSCLE RELAXATION
- NO EFFECT ON CVS AND CEREBRAL BLOOD FLOW
- SAFEST ANESTHETIC

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- CAN CAUSE DIFFUSION HYPOXIA, PNEUMOTHORAX, PRESSURE IN SINUSES

QUESTION

A patient receives nitrous oxide for a short dental procedure. The anesthetist explains one of its important pharmacological properties.

A. It is a potent anesthetic used alone for major surgery.

B. It provides good analgesia but weak anesthesia.

C. It is metabolized extensively in the liver.

D. It commonly causes severe hepatotoxicity.

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ADVERSE EFFECTS OF INHALATION ANESTHETICS

1) Hepatotoxicity
2) Nephrotoxicity
3) Malignant Hyperthermia
4) Mutagenecity
5) Carcinogenicity

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MALIGNANT HYPERTHERMIA

IN SUSCEPTIBLE PATIENTS' EXPOSURE TO HALOGENATED HYDROCARBON ANESTHETICS OR SUCCINYLCHOLINE MAY INDUCE MALIGNANT HYPERTHERMIA, A RARE LIFE-THREATENING CONDITION

S/S

HYPERTHERMIA

MUSCLE RIGIDITY

CIRCULATORY COLLAPE

DEATH

UNCONTROLLED RELEASE OF CALCIUM

PATIENTS AT RISK

BURN VICTIMS

MUSCULAR DYSTROPHY

MYOPATHY

MYOTONIA

OSTEOGENESIS IMPERFECTA

AUTOSOMAL DOMINENT DISORDERS

TREATMENT

DENTROLENE (BLOCKS CALCIUM RELEASE REDUCES HEAT PRODUCTION AND RELAXING MUSCLE TONE)

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QUESTION

A young man develops high fever, muscle rigidity, and hypercapnia after receiving a halogenated inhalational anesthetic. The team suspects malignant hyperthermia.

A. Family history of malignant hyperthermia

B. Controlled hypertension

C. Diabetes mellitus

D. Iron-deficiency anemia

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THANK YOU